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# Study of Ground Water Quality on the Basis of Heavy Metal Pollution Index and its Impact on Human Health and Agriculture in East Kachchh .Gujarat.

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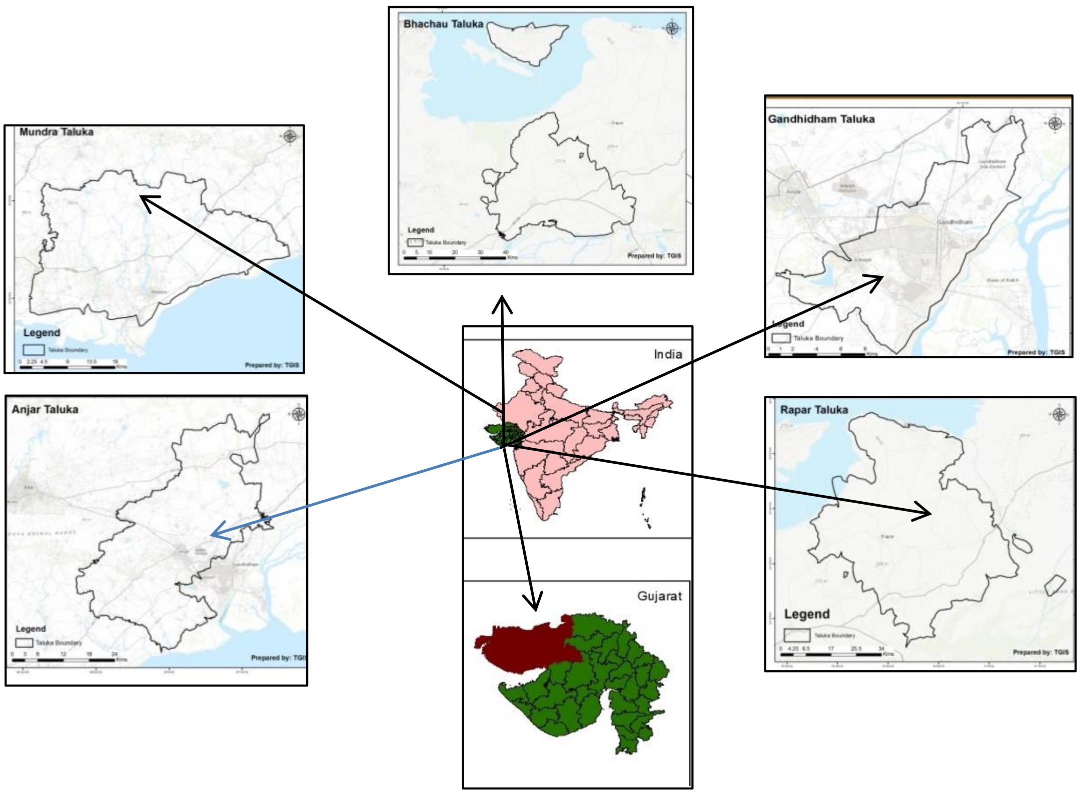
**Abstract**

Heavy metal contamination in Groundwater is a serious issue related to rapid industrialization and its health effects on the human population, plants and animals. Although some metals like Iron, Manganese, Magnesium, calcium and fluoride are essential up to certain permissible limits, more than recommended concentration the cause serious health effects. In this study, groundwater samples are collected from industrial peripheral regions, after Monsoon 2021 and analyzed using an atomic Absorption Spectrophotometer, Flame photometer. It is found that out of 75 sample stations in five talukas of East Kachchh 75 % of samples have Heavy Metal Index > 100 test water is contaminated with heavy metal, and 15 % of groundwater samples have HPI <100 indicating that they are free from heavy metal contamination and suitable for drinking.

**Key Words** - Heavy netals, East Kachchh , Groundwater . Heavy Metal Pollution Index (HMPI)

## 1. Study area



## 2. Introduction

Worldwide, it is estimated that the industry is responsible for dumping 300-400 million tons of heavy metals, solvents, toxic sludge, and other waste into waters each year (UNEP, 2010). (Pawar V. and Gawade S., 2016)

The industrialization has become an important parameter to measure the development of a country's economy through the establishment of industrial sectors. However, the waste or by-products discharged from them are severely disastrous to the environment and consists of various kind of contaminant which contaminate the surface water, groundwater and soil. There are a number of reasons the waste is not safely treated. One of the reasons is mainly due to the lack of highly efficient and economic treatment technology. The surface and groundwater resources are steadily declining because of the increase in population, industrial growth, pollution by various human, agricultural and industrial wastes and unexpected climate change. (Gajendran C, 2011).

In the Kachchh region, Soils are Loamy (medium textured), Saline due to the large coastline and Saline desert. The salt content is very high with sodium chloride as the dominant soil leads to the leaching of soluble sodium salts in Groundwater. The Great Rann of Kutch is a salt marsh in the Thar Desert in the Kutch District of Gujarat, India. It is about 7500 Km2 in the area and is reputed to be one of the largest salt deserts in the world where the temperature is 440 during hot summer and even rises to 500, highly seasonal rainfall and excess evaporation rate leads to dry ran before next water availability.

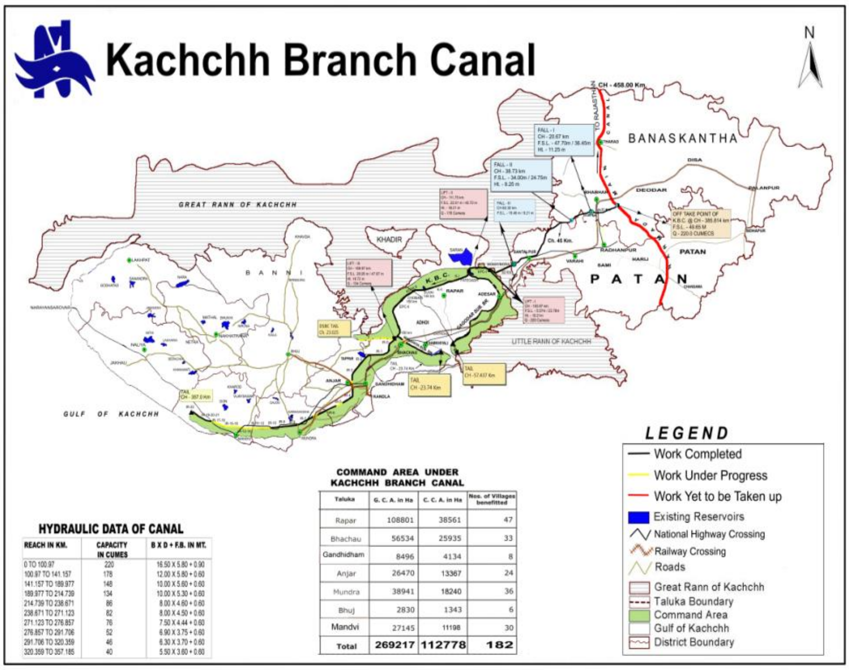
The average rainfall of Kachchh is 346 mm but in the last few years it has considerably increased thus Water table increase is seen along with that there is the Narmada Branch canalprovides surface water to the Kachchh region. According to the report by the metrological department

"The incidence of heavy rainfall has kept increasing over the last 40 years in the coastal areas of the state and one of the key reasons for this is global warming and climate change." Another important reason for this change is the fact that, in the past decade, Gujarat has gotten most of its rain from systems forming over the Arabian Sea. (source- Indian Metrology Department -2010)

The coastline of Kachchh is 406 Km. 75% of Salt production is in Gandhidham and Bhachau Taluka .salt concentration is too high in air and soil so as in Groundwater due to washing of soluble salts. Saltwater intrusion can be a problem in coastal areas where rates of groundwater pumping are high enough to cause seawater to invade freshwater aquifers.

Kachchh has become an industrial hub after severe earthquake on 26th January 2001, as it was declared a special economic zone for the industrial setup to lure various industrial units to establish and contribute to the development of Kachchh which was shattered due to the devastations of the earthquake. Kachchh is having a source of the Narmada canal but still, that is limited to certain regions and groundwater supplement source is fulfilling the increasing demand for water supply due to the population. In an area like Kachchh, the study region is semi-arid having a wide range of temperature differences in a day Narmada water canal system will prove to be a boon for remote areas of Kachchh.

**Figure 1- Proposed Narmada Water Canal for Kachchh**




(Source- Narmada Nigam , Gandhidham )

The Command area under the Kachchh branch water canal will directly benefit 182 villages of Kachchh. In east Kachchh, Narmada main Branch canal will fulfil the water requirements of five talukas of east Kachchh as Rapar-47, Bhachau-33, Gandhidham- 8, Anjar-24 and Mundra 36 villages. Peripheral areas will be having sub-branch canals with surplus water.

**Table 1- Desired level of various heavy metals in Potable Water (Guidelines of WHO and IS: 10500-1991)**

| Sr. No | Parameter | Desired limit In ppm | Undesirable effect outside the desirable limit | Permitted limit in absence of other source |
|---|---|---|---|---|
| 1 | Iron as Fe, | 0.03 | Beyond this limit taste/appearance are affected, has adverse effect on domestic uses and water supply structures, and promotes iron bacteria. | 1.0 |
| 2 | Calcium as Ca | 75 | Encrustation in water supply structure and adverse effects on domestic use | 200 |
| 3 | Copper as Cu | 0.05 | Astringent taste, discoloration and corrosion of pipes, fitting and utensils will be caused beyond this | 1.5 |
| 4 | Manganese as Mn, | 0.1 | Beyond this limit taste/appearance are affected, has adverse effect on domestic uses and water supply structures | 0.3 |
| 5 | Fluoride | 1 | Fluoride may be kept as low as possible. High fluoride may cause fluorosis | 1.5 |
| 6 | Cadmium as Cd | 0.01 | Beyond this, the water becomes toxic | No relaxation |

| | | | | |
|---|---|---|---|---|
| 7 | Lead as Pb | 0.05 | Beyond this, the water becomes toxic | No relaxation |
| 8 | Zinc as Zn, | 5 | Beyond this limit it can cause astringent taste and an opalescence in water | 15 |
| 9 | Sodium as Na | 200 | | |
| 10 | Potassium as $K_2O$ | 1 | | 4 |

## 3. Effect of Heavy metals in Water on Human being

Environment pollution particularly from hazardous heavy metals and minerals (fluoride arsenic salts) is an important societal problem. Many of these elements being stable are bio-accumulative, and deriving their safe limits is very difficult. Also, the toxicity of metals depends largely on their chemical form and oxidation state. The data of the central pollution control board (CPCB) shows that Gujarat Maharashtra and Andra Pradesh contribute to around 80% of hazardous waste including heavy metals. Central Pollution

Some elements like Fe, Zn, Cu, Co, Cr, Mn, and Ni, are needed in small quantities for human metabolism but may be toxic at higher levels. Others like lead, mercury, cadmium, arsenic, etc. have no beneficial role and are positively toxic.

**3.1 Fluorid**e- Small amounts of fluoride help to prevent dental caries, but excess is harmful. The toxicity of these is of considerable concern in India because of their environmental burden. Natural sources contribute to the bulk of the environmental load of fluoride and arsenic.

3.**2 LEAD (Pb)** lead is chemically similar to calcium, the body handles it like calcium. In the body lead is distributed throughout bone, teeth, liver, lung, brain and spleen; bone being the major accumulator. Lead can cross the blood-brain barrier as well as the placental barrier. (Akyüz, M., and Çabuk, H. (2010) Excretion occurs through urine and feces. In general, Pb is excreted very slowly from the body. It's biological half-life estimated at 10 years, facilitates accumulation in the body.

3.**3 COPPER (Cu)-** Copper pollution in the environment comes from human activities and natural sources. Manufacturing companies dealing with copper in the production of metal, electrical appliances, pesticides, fungicides and other products containing copper often release contaminated water into the drainage system and may affect humans consuming unwashed fruits or vegetables sprayed with such pesticides. Copper salts are sometimes especially added to water reservoirs to inhibit the growth of algae. Copper salts are extensively used in agricultural pesticide spray. (Hem J.D., 1989)

An excess of copper in the water has adverse effects on aquatic life, with damage to freshwater organisms such as fish. Copper damages the kidneys, nervous systems, and livers of most water creatures. Humans require low levels of copper to maintain good health. When the metal builds up to high levels in the body, health is affected.

3.**4 CADMIUM (Cd)-** Cadmium is widely used in industrial processes, e.g.: as an anticorrosive agent, as a stabilizer in PVC products, as a colour pigment, as a neutron-absorber in nuclear power plants, and in the fabrication of nickel-cadmium batteries. Phosphate fertilizers also show a big cadmium load. (Hem J.D., 1989), chronic anemia can be caused by prolonged exposure to drinking water contaminated with cadmium (Cd). Under such circumstances, accumulation of Cd is manifested in the kidney, resulting in cancer and cardiovascular diseases.

**3.5. IRON (Fe) -** Elemental iron is rarely found in nature, as the iron ions Fe2+ and Fe3+ readily combine with oxygen- and sulfur-containing compounds to form oxides, hydroxides, carbonates, and sulfides. Iron is most commonly found in nature in the form of its oxides. Concentrations of iron in drinking water are normally less than 0.3 mg/litre but may be higher in countries where various iron salts are used as coagulating agents in water-treatment plants and where cast iron, steel, and galvanized iron pipes are used for water distribution. No health-based guideline value for iron is proposed

**3.6 Manganese (Mn)** – Manganese is not an essential constituent of any of the more common silicate rock minerals, but it can substitute iron, magnesium, or calcium in silicate structures. Many igneous and metamorphic minerals contain divalent manganese as a minor constituent .when divalent manganese is released into the aqueous solution during weathering it remains stable towards oxidation than ferrous iron. (Hem J.D., 1989)

**3.7 Potassium-** is an essential element for both plants and animals. Maintenance of optimum soil fertility entails providing a supply of available potassium. The element is present in plant material and is lost from agricultural soil by crop harvesting and removal as well as by leaching and runoff acting on organic residues. (NGWA,2010).

**3.8 Sodium** is the most abundant member of the alkali metal group. In igneous rock, sodium is more abundant than potassium but in sediments, sodium is much less than the amount of sodium held in evaporating sediments and in solutions in the oceans are an important part of the total as the study area is close to the sea coast and the crude salt is kept in open areas near the purification plants. The presence of sodium in the air cannot be ignored when is washed with rain water or sedimented even too far off is also considered as the sodium in the soil so as in the groundwater. (NGWA- the Groundwater Association, 2010)

**3.9 Zinc** is used extensively as a white pigment (zinc oxide) in paint and rubber. These applications tend to disperse the element widely in the environment, and its availability for solution in water has been greatly enhanced by industrial civilization. (Hem J.D.,1989) Zinc is an essential element in plant and animal metabolism but water is not a significant source component of a dietary sense.

## 4. Methodology

To examine the extent of the contamination by toxic metals leached from tailings, 75 groundwater, samples were collected and analysed from the studied area. post monsoon seasons. Samples were collected and preserved in a pre-cleaned, sterilized polyethylene bottles. After collection, the samples were transported to the laboratory within 24 hrs and subjected to analysis for various parameters such as pH, electrical conductivity, salinity, total dissolved solids, total hardness, calcium hardness, magnesium hardness, fluoride, potassium, sulphate, chlorides, magnesium, nitrate, phosphate, sodium and heavy metals – Lead, Cadmium, Zinc, Iron, Nickel, Copper and Manganese samples were estimated in the Laboratory by using Standard Procedures. Fine grade chemicals were used throughout the study. All the reagents and standards required for the study was prepared using Millipore water.

Table -2

| | | |
|---|---|---|
| 1 | Sodium and potassium | Flame photometer |
| 2 | Fluoride | fluoride kit (in situ) |
| 4 | Zinc, Copper, Iron ,Cobalt, Chromium, Lead, Manganese, | Atomic Absorption spectrophotometer(Model- |

5. **Statistical Calculation- Heavy Metal Pollution Index - Mohen et al. (1996)**

$$HPI = \frac{\sum_{i=1}^{n} W_i \, Q_i}{\sum_{i=1}^{n} W_i}$$

$W_i$=Unit weight of the $i^{th}$ Parameters

$Q_i$=Sub-Index value of the $i^{th}$ Parameters

$$Q_i = \sum_{i=1}^{n} \frac{(M_i\,(-)\,I_i)}{(S_i - I_i)} * 100$$

$M_i$=Monitored value of heavy metal of the $i^{th}$ parameters.

$$W_i = \frac{K}{S_i}$$

$I_i$=Ideal value of the $i^{th}$ parameters

$S_i$= Standard value of the $i^{th}$ parameters

Where K=Constant=1 and $S_i$=Standard Permissible limit Value of the $i^{th}$ parameters

## 6. Results and conclusion

**Heavy metal pollution Index calculation**

### 6.1.1 Table -3 Gandhidham (HMPI)

| sr no | Parameter | Ii (ppb) | Si (ppb) | Station 1 | Station 2 | Station 3 | Station 4 | Station 5 | Station 6 | Station 7 | Station 8 | Station 9 | Station 10 | Station 11 | Station 12 | Station 13 | Station 14 | Station 15 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Average HPI for all heavy metals | | | | 349.390 | 167.544 | 178.932 | 209.357 | 10.045 | 160.025 | 273.966 | 169.285 | 319.718 | 178.077 | 242.417 | 120.559 | 521.252 | 271.145 | 214.628 |
| 1 | Cd | 0 | 3 | 169.850 | 107.269 | 98.345 | 187.561 | 6.796 | 80.383 | 133.972 | 89.398 | 160.926 | 98.345 | 143.032 | 80.452 | 107.269 | 151.979 | 116.216 |
| 2 | F | 1000 | 1500 | 0.272 | 0.272 | 0.272 | 0.270 | 0.041 | 0.272 | 0.272 | 0.272 | 0.272 | 0.272 | 0.272 | 0.272 | 0.272 | 0.272 | 0.272 |
| 3 | Zn | 5000 | 15000 | 0.007 | 0.007 | 0.007 | 0.007 | 0.002 | 0.007 | 0.007 | 0.007 | 0.007 | 0.007 | 0.007 | 0.007 | 0.007 | 0.007 | 0.007 |
| 4 | Pb | 0 | 10 | 117.902 | 39.238 | 39.299 | 16.881 | 2.036 | 58.857 | 78.476 | 58.951 | 117.902 | 58.951 | 78.603 | 39.301 | 393.013 | 98.252 | 78.603 |
| 5 | Na | 0 | 200000 | 0.002 | 0.003 | 0.003 | 0.000 | 0.000 | 0.000 | 0.002 | 0.002 | 0.002 | 0.002 | 0.002 | 0.002 | 0.002 | 0.002 | 0.002 |
| 6 | K | 1000 | 4000 | 0.468 | 0.215 | 0.346 | 0.017 | 0.007 | 0.170 | 0.310 | 0.165 | 0.187 | 0.184 | 0.155 | 0.181 | 0.177 | 0.189 | 0.324 |
| 7 | Ni | 0 | 20 | 60.492 | 20.130 | 40.311 | 4.220 | 1.018 | 20.171 | 60.492 | 20.173 | 40.345 | 20.173 | 20.173 | 0.000 | 20.173 | 20.173 | 18.877 |
| 8 | Mn | 100 | 300 | 0.327 | 0.331 | 0.294 | 0.312 | 0.102 | 0.108 | 0.308 | 0.313 | 0.011 | 0.137 | 0.109 | 0.294 | 0.285 | 0.234 | 0.109 |
| 9 | Cu | 50 | 1500 | 0.004 | 0.004 | 0.005 | 0.004 | 0.014 | 0.004 | 0.004 | 0.004 | 0.004 | 0.004 | 0.004 | 0.004 | 0.004 | 0.004 | 0.004 |
| 10 | Fe | 300 | 1000 | 0.066 | 0.075 | 0.050 | 0.085 | 0.029 | 0.051 | 0.122 | 0.000 | 0.061 | 0.002 | 0.060 | 0.047 | 0.050 | 0.032 | 0.215 |

Out of 15 samples of Taluka Gandhidham only one sample is having Heavy metal toxicity less than 100 which indicate that it is free from metal toxicity 6 samples are free from Cadmium and station no.1 has toxicity of Lead and Cadmium.

### 6.1.2 Table -4 Anjar (()HMPI)

| sr no | Parameter | Ii (ppb) | Si (ppb) | Station 1 | Station 2 | Station 3 | Station 4 | Station 5 | Station 6 | Station 7 | Station 8 | Station 9 | Station 10 | Station 11 | Station 12 | Station 13 | Station 14 | Station 15 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Average HPI for all heavy metals | | | | 534.000 | 505.053 | 267.572 | 355.555 | 308.131 | 173.626 | 50.351 | 134.153 | 1681.052 | 10.045 | 179.735 | 0.959 | 0.901 | 159.673 | 336.536 |
| 1 | Cd | 0 | 3 | 408.729 | 317.900 | 249.779 | 313.359 | 283.840 | 158.950 | 22.707 | 124.889 | 1385.137 | 6.803 | 84.017 | 0.182 | 0.091 | 90.829 | 204.365 |
| 2 | F | 1000 | 1500 | 0.093 | 0.087 | 0.016 | 0.076 | 0.338 | 0.075 | 0.624 | 0.132 | 0.180 | 0.041 | 0.477 | 0.272 | 0.272 | 0.673 | 0.063 |
| 3 | Zn | 5000 | 15000 | 0.007 | 0.007 | 0.007 | 0.006 | 0.006 | 0.007 | 0.007 | 0.007 | 0.007 | 0.002 | 0.007 | 0.007 | 0.007 | 0.006 | 0.007 |
| 4 | Pb | 0 | 10 | 14.306 | 10.218 | 8.175 | 14.306 | 10.218 | 6.131 | 16.349 | 4.087 | 292.241 | 2.026 | 94.008 | 0.053 | 0.082 | 67.440 | 130.794 |
| 5 | Na | 0 | 200000 | 0.001 | 0.002 | 0.003 | 0.004 | 0.004 | 0.002 | 0.005 | 0.001 | 0.002 | 0.000 | 0.003 | 0.000 | 0.000 | 0.005 | 0.004 |
| 6 | K | 1000 | 4000 | 0.290 | 0.029 | 0.075 | 0.043 | 1.209 | 0.170 | 0.068 | 0.058 | 0.012 | 0.007 | 0.068 | 0.017 | 0.017 | 0.065 | 0.483 |
| 7 | Ni | 0 | 20 | 110.357 | 12.262 | 9.196 | 27.589 | 12.262 | 7.664 | 10.218 | 4.598 | 3.065 | 1.021 | 0.613 | 0.000 | 0.001 | 0.511 | 0.766 |
| 8 | Mn | 100 | 300 | 0.153 | 0.259 | 0.259 | 0.095 | 0.177 | 0.565 | 0.300 | 0.300 | 0.259 | 0.102 | 0.467 | 0.336 | 0.340 | 0.080 | 0.050 |
| 9 | Cu | 50 | 1500 | 0.031 | 0.008 | 0.002 | 0.027 | 0.005 | 0.002 | 0.008 | 0.002 | 0.003 | 0.014 | 0.004 | 0.005 | 0.005 | 0.004 | 0.004 |
| 10 | Fe | 300 | 1000 | 0.035 | 164.280 | 0.061 | 0.051 | 0.072 | 0.061 | 0.064 | 0.080 | 0.146 | 0.029 | 0.073 | 0.087 | 0.088 | 0.060 | 0.001 |

In Anjar Taluka. In the analysis of post-monsoon season 2021 (October to December 2021) it is found that the average Heavy Metal Index of all the heavy metals is in the range of 10.04 - 521.252 out of 15 stations station no.5 is free from the Heavy Metal pollution in which major Heavy metal contributor is Cadmium In sample 1

Cd and Ni are the major contributor for the Metal toxicity and in station no. 9 and 15, Pb is also found to be the major polluting agent

### 6.1.3 Table -4 Mundra (HMPI)

| sr no | Parameter | Ii (ppb) | Si (ppb) | Station 1 | Station 2 | Station 3 | Station 4 | Station 5 | Station 6 | Station 7 | Station 8 | Station 9 | Station 10 | Station 11 | Station 12 | Station 13 | Station 14 | Station 15 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Average HPI for all heavy metals | | | | 310.6406 | 382.8308 | 296.5499 | 355.5554 | 569.1642 | 173.6258 | 292.4302 | 693.7798 | 258.3866 | 19.17018 | 356.3008 | 199.6430 | 239.09015 | 80.3104 | 305.79857 |
| 1 | Cd | 0 | 3 | 204.3645 | 225.4595 | 211.1767 | 313.3590 | 402.5982 | 177.1386 | 193.2607 | 96.6190 | 96.6190 | 12.7160 | 225.4822 | 129.4309 | 158.9502 | 22.7072 | 193.2614 |
| 2 | F | 1000 | 1500 | 0.0000 | 0.0000 | 0.1362 | 0.0763 | 0.1362 | 0.8175 | 0.6812 | 0.0000 | 0.1362 | 0.0136 | 0.1362 | 0.2725 | 0.1362 | 0.6730 | 0.0000 |
| 3 | Zn | 5000 | 15000 | 0.0065 | 0.0066 | 0.0065 | 0.0063 | 0.0065 | 0.0067 | 0.0067 | 0.0067 | 0.0066 | 0.0007 | 0.0067 | 0.0067 | 0.0068 | 0.0065 | 0.0068 |
| 4 | Pb | 0 | 10 | 59.2371 | 88.8536 | 59.2371 | 14.3055 | 148.0907 | 29.6186 | 88.8536 | 592.3711 | 156.9520 | 5.9237 | 118.4742 | 59.2371 | 59.2657 | 47.0038 | 102.1826 |
| 5 | Na | 0 | 200000 | 0.0036 | 0.0029 | 0.0053 | 0.0040 | 0.0053 | 0.0033 | 0.0037 | 0.0061 | 0.0032 | 0.0003 | 0.0038 | 0.0005 | 0.0009 | 0.0039 | 0.0032 |
| 6 | K | 1000 | 4000 | 0.5313 | 0.0238 | 0.0971 | 0.0426 | 0.1448 | 0.0630 | 0.0800 | 0.1890 | 0.0324 | 0.0385 | 0.1226 | 0.1107 | 0.0141 | 0.0647 | 0.0324 |
| 7 | Ni | 0 | 20 | 45.9820 | 68.1198 | 25.5456 | 27.5892 | 17.8819 | 6.8120 | 8.5154 | 4.2574 | 4.2574 | 0.4257 | 11.9216 | 10.2182 | 20.4365 | 9.7073 | 10.2183 |
| 8 | Mn | 100 | 300 | 0.3304 | 0.3297 | 0.2936 | 0.0954 | 0.2555 | 0.2936 | 0.4632 | 0.2759 | 0.3100 | 0.0358 | 0.1124 | 0.2943 | 0.2602 | 0.0800 | 0.0545 |
| 9 | Cu | 50 | 1500 | 0.0040 | 0.0043 | 0.0045 | 0.0267 | 0.0042 | 0.0040 | 0.0041 | 0.0043 | 0.0040 | 0.0004 | 0.0043 | 0.0043 | 0.0043 | 0.0044 | 0.0038 |
| 10 | Fe | 300 | 1000 | 0.1810 | 0.0304 | 0.0473 | 0.0505 | 0.0410 | 0.0627 | 0.5615 | 0.0503 | 0.0658 | 0.0155 | 0.0366 | 0.0677 | 0.0152 | 0.0596 | 0.0366 |

Analysis results i[n] … on no 10 and 14 are free fr[o] … hile Pb toxicity is found i … hore than 100.

### 6.1.4 Table no-5 Rapar (HMPI)

| sr no | Parameter | Ii (ppb) | Si (ppb) | Station 1 | Station 2 | Station 3 | Station 4 | Station 5 | Station 6 | Station 7 | Station 8 | Station 9 | Station 10 | Station 11 | Station 12 | Station 13 | Station 14 | Station 15 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Average HPI for all heavy metals | | | | 258.019 | 505.608 | 311.194 | 322.447 | 296.602 | 175.081 | 569.260 | 214.437 | 1091.643 | 16.074 | 179.735 | 191.800 | 357.049 | 200.409 | 240.126 |
| 1 | Cd | 0 | 3 | 227.072 | 317.900 | 204.365 | 225.460 | 211.177 | 158.950 | 402.598 | 177.139 | 193.261 | 9.662 | 84.017 | 127.16016 | 225.482 | 129.431 | 158.951 |
| 2 | F | 1000 | 1500 | 0.000 | 0.136 | 0.272 | 0.817 | 0.136 | 0.272 | 0.272 | 0.272 | 0.136 | 0.014 | 0.477 | 0 | 0.136 | 0.000 | 0.272 |
| 3 | Zn | 5000 | 15000 | 0.007 | 0.007 | 0.007 | 0.006 | 0.006 | 0.007 | 0.007 | 0.007 | 0.007 | 0.001 | 0.007 | 0.0067767 | 0.007 | 0.006 | 0.007 |
| 4 | Pb | 0 | 10 | 20.436 | 10.218 | 59.237 | 88.872 | 59.237 | 6.131 | 148.091 | 29.619 | 888.536 | 5.924 | 94.008 | 59.237107 | 118.474 | 59.237 | 59.266 |
| 5 | Na | 0 | 200000 | 0.001 | 0.001 | 0.003 | 0.003 | 0.008 | 0.001 | 0.000 | 0.000 | 0.000 | 0.000 | 0.003 | 0.0023298 | 0.003 | 0.002 | 0.003 |
| 6 | K | 1000 | 4000 | 0.284 | 0.536 | 0.812 | 0.112 | 0.146 | 1.429 | 0.109 | 0.228 | 0.158 | 0.016 | 0.068 | 0.6199058 | 0.872 | 1.148 | 0.911 |
| 7 | Ni | 0 | 20 | 10.218 | 12.262 | 45.982 | 6.812 | 25.546 | 7.664 | 17.882 | 6.812 | 8.515 | 0.426 | 0.613 | 4.2574244 | 11.922 | 10.218 | 20.437 |
| 8 | Mn | 100 | 300 | 0.000 | 0.259 | 0.330 | 0.330 | 0.294 | 0.565 | 0.255 | 0.294 | 0.463 | 0.028 | 0.467 | 0.357638 | 0.112 | 0.294 | 0.260 |
| 9 | Cu | 50 | 1500 | 0.000 | 0.008 | 0.004 | 0.004 | 0.005 | 0.002 | 0.004 | 0.004 | 0.004 | 0.000 | 0.004 | 0.0043145 | 0.004 | 0.004 | 0.004 |
| 10 | Fe | 300 | 1000 | 0.000 | 164.280 | 0.181 | 0.030 | 0.047 | 0.061 | 0.041 | 0.063 | 0.562 | 0.005 | 0.073 | 0.154671 | 0.037 | 0.068 | 0.015 |

having Cd toxicity while in others that is station no. 7 and 9 key element is are found to be with Pb also that means most of the water is not fit for drinking because of heavy metal toxicity which is more than 100.

### 6.1.5 Table 6 Bhachau (HMPI

| sr no | Parameter | Ii (ppb) | Si (ppb) | Station 1 | Station 2 | Station 3 | Station 4 | Station 5 | Station 6 | Station 7 | Station 8 | Station 9 | Station 10 | Station 11 | Station 12 | Station 13 | Station 14 | Station 15 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Average HPI for all heavy metals | | | | 534.114 | 505.546 | 267.7537 | 355.893 | 30.689 | 173.727 | 50.001 | 134.348 | 434.651 | 276.697 | 179.750 | 239.911 | 173.566 | 159.052 | 37.284 |
| 1 | Cd | 0 | 3 | 408.729 | 317.900 | 249.779 | 313.359 | 28.384 | 158.950 | 22.707 | 124.889 | 138.514 | 90.829 | 90.829 | 181.657 | 90.829 | 90.829 | 36.332 |
| 2 | F | 1000 | 1500 | 0.272 | 0.954 | 0.272 | 0.272 | 0.014 | 0.272 | 0.272 | 0.272 | 0.272 | 0.272 | 0.272 | 0.272 | 0.272 | 0.000 | 0.272 |
| 3 | Zn | 5000 | 15000 | 0.007 | 0.007 | 0.007 | 0.006 | 0.001 | 0.007 | 0.007 | 0.007 | 0.007 | 0.007 | 0.007 | 0.007 | 0.007 | 0.006 | 0.007 |
| 4 | Pb | 0 | 10 | 14.306 | 10.218 | 8.175 | 14.306 | 1.022 | 6.131 | 16.349 | 4.087 | 292.241 | 175.754 | 175.754 | 53.135 | 81.746 | 67.440 | 0.000 |
| 5 | Na | 0 | 200000 | 0.003 | 0.003 | 0.001 | 0.001 | 0.000 | 0.000 | 0.000 | 0.000 | 0.002 | 0.001 | 0.001 | 0.002 | 0.000 | 0.001 | 0.001 |
| 6 | K | 1000 | 4000 | 0.221 | 0.174 | 0.003 | 0.187 | 0.017 | 0.075 | 0.075 | 0.112 | 0.141 | 0.198 | 0.198 | 0.158 | 0.092 | 0.121 | 0.201 |
| 7 | Ni | 0 | 20 | 110.357 | 12.262 | 9.196 | 27.589 | 1.226 | 7.664 | 10.218 | 4.598 | 3.065 | 9.196 | 9.196 | 0.153 | 0.511 | 0.511 | 0.204 |
| 8 | Mn | 100 | 300 | 0.153 | -0.259 | 0.259 | 0.095 | 0.018 | 0.565 | 0.300 | 0.300 | 0.259 | 0.259 | 0.259 | 4.407 | 0.077 | 0.080 | 0.132 |
| 9 | Cu | 50 | 1500 | 0.031 | 0.008 | 0.002 | 0.027 | 0.000 | 0.002 | 0.008 | 0.002 | 0.003 | 0.004 | 0.004 | 0.004 | 0.004 | 0.004 | 0.005 |
| 10 | Fe | 300 | 1000 | 0.035 | 164.280 | 0.061 | 0.051 | 0.007 | 0.061 | 0.064 | 0.080 | 0.146 | 0.178 | 0.178 | 0.114 | 0.028 | 0.060 | 0.130 |

In Taluka Bhachau only 3 stations are free from the HMP index. Pollution contributor is Cd the range of Cd, HMPI is found to be from 22.707- 408.729. Only 7 samples are free from cd pollution where HMPI < 100 and Ni Concentration is high in station no. 1 while Pb along with Cd is found in station no. 9 and 11 Index is more than 100.

### 6.2 Conclusion

During the post Monsoon 2021 analysis it is found that in East Kachchh there are more than 750 Industrial units working . Out of five talukas 4 talukas are industrially saturated except Rapar , although the efficiency is less because of water salinity due to recycling and reuse of water . Calculating average Heavy metal Index and Individual HPI of each metal it is found that Cadmium is a main contributor in Metal toxicity somewhere Lead, Nickel and Sodium are also present with Cadmium.

In the study region soil is loamy and porous. Percolation of surface water to ground water plays major role in increasing concentration of pollutants. In some areas soil is also analysed and found that the level of metals are high but remain persistent in the upper layers in non-soluble forms , the metals having ionic constituents and if soluble , are leaching to the ground water table. According to soil sample analysis in this region it is found that although Metal concentration in some of the stations is high due to natural factors rather than industrial factors . Leaching is also found restricted only to upper strata of soil. Metals which form soluble complexes percolate along with the water as soil is porous.

To reduce the concentration of metals in ground water is possible by frequent irrigation as Narmada branch canal and increase in downpour due to climate change. One of the groups of salt tolerant plants is 'Halophyte'. These plants have the ability to not only to tolerate excess salt along with normal growth in saline environmental conditions.

Leaching may reduce salinity levels in the absence of artificial drains when there is sufficient natural drainage Salts are most efficiently leached from the soil profile under higher frequency irrigation. As Narmada Water canal is available in Kachchh and 182 villages are directly benefitted by Kachchh Branch Canal, will reduce salinity is soil profile and Ground water as well.

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In the study, the region's soil is loamy and porous. Percolation of surface water to groundwater plays a major role in increasing the concentration of pollutants. In some areas soil is also analysed and found that the level of metals are high but remain persistent in the upper layers in non-soluble forms, the metals having ionic constituents and if soluble, are leaching to the groundwater table. According to soil sample analysis in this region, it is found that although Metal concentration in some of the stations is high due to natural factors rather than industrial factors. Leaching is also found restricted only to the upper strata of soil. Metals that form soluble complexes percolate along with the water as the soil is porous.

Reducing the concentration of metals in groundwater is possible by frequent irrigation as Narmada branch canal and increase in downpours due to climate change. One of the groups of salt-tolerant plants is 'Halophyte'. These plants have the ability not only to tolerate excess salt along with normal growth in saline environmental conditions.

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